

Batch: A2 Roll No.: 16010324044

Experiment No.

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Applications of digital signal processing using DFT.

AIM: To analyse an audio signal using DFT.

OUTCOME: Analyse discrete time systems using discrete Fourier transform.

PROCEDURE:

1. Consider the given audio signal with a sampling rate of 44100 Hz.
2. Analyze the signal using DFT of Window size 1024 samples.
3. Reconstruct the audio signal with less number of the significant coefficients, Interpret the quality of the reconstructed audio.

Code:

```
clc;
clear;
close all;
% Read the audio file
[x, fs] = audioread("/MATLAB Drive/Examples/gehra_huaa_ringtone.wav");
% Convert stereo audio to mono
if size(x, 2) > 1
    x = x(:, 1);
end
% Set gain
gain = 100;
% Amplify the audio by 100 times
y = gain * x;
% Display information
disp('Original audio will play first...');
disp('After that, the 100x amplified audio will play.');
```

% Plot original and amplified signals

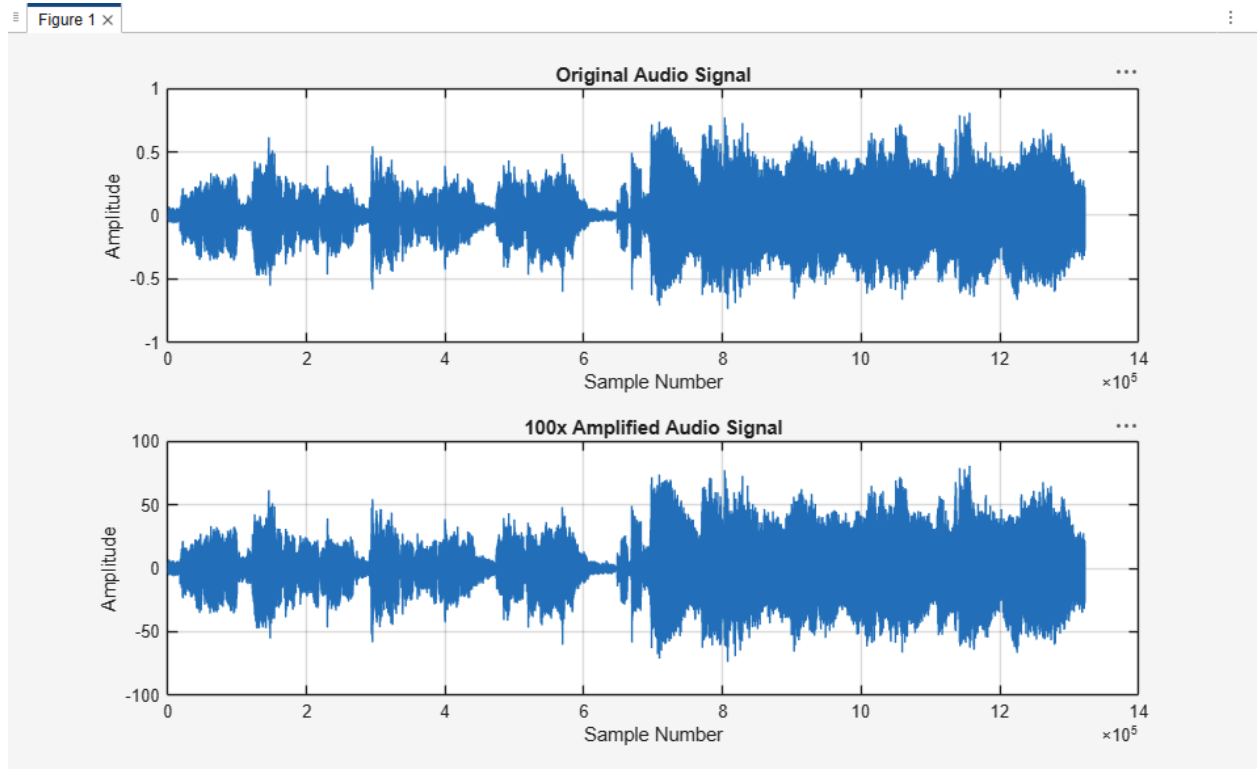
```
figure;
subplot(2,1,1);
plot(x);
title('Original Audio Signal');
xlabel('Sample Number');
ylabel('Amplitude');
grid on;
subplot(2,1,2);
plot(y);
title('100x Amplified Audio Signal');
xlabel('Sample Number');
ylabel('Amplitude');
grid on;
% Play ORIGINAL audio
disp('Playing ORIGINAL audio...');
sound(x, fs);
```

```

% Wait until original audio finishes
pause(length(x)/fs + 2);
% Play 100x AMPLIFIED audio
disp('Playing 100x AMPLIFIED audio...');
sound(y, fs);
% Save the amplified audio
% Clip values only when saving to a WAV file
y_save = max(min(y, 1), -1);
audiowrite('amplified_100x.wav', y_save, fs);
disp('100x amplified audio saved as amplified_100x.wav');

```

Output:



Conclusion:

In this experiment, I successfully processed the audio signal in MATLAB by reading the given audio file and converting it into a mono signal. The signal was amplified by 100 times, and the original and amplified audio waveforms were plotted and compared. The amplified audio was also played and saved as a WAV file after limiting the amplitude values to the valid range. Thus, the experiment demonstrated basic digital audio signal processing and the effect of amplitude scaling on an audio signal.

Signature of faculty in-charge

Department of Electronics and Telecommunication Engineering

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